



Oral Surgery

Third stage

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Complications of Exodontia (part 1)

- Complications can arise during the procedure of extraction or may manifest themselves sometime following the extraction.
- All these complications arise from **error in judgment, misuse of instruments, exertion of extensive force or from anatomic causes.**

1- Failure to remove the tooth with either forceps or elevator.

Failure to remove or even move the tooth after applying a reasonable amount of force, need further clinical and radiological evaluation, because the tooth may be need surgical extraction.

2- Fracture: of

A. Crowns and roots.

This is the most common complication during tooth extraction.

The factors that may lead to this complication may be **classified into three groups:**

➤ Factors related to the tooth itself, include:

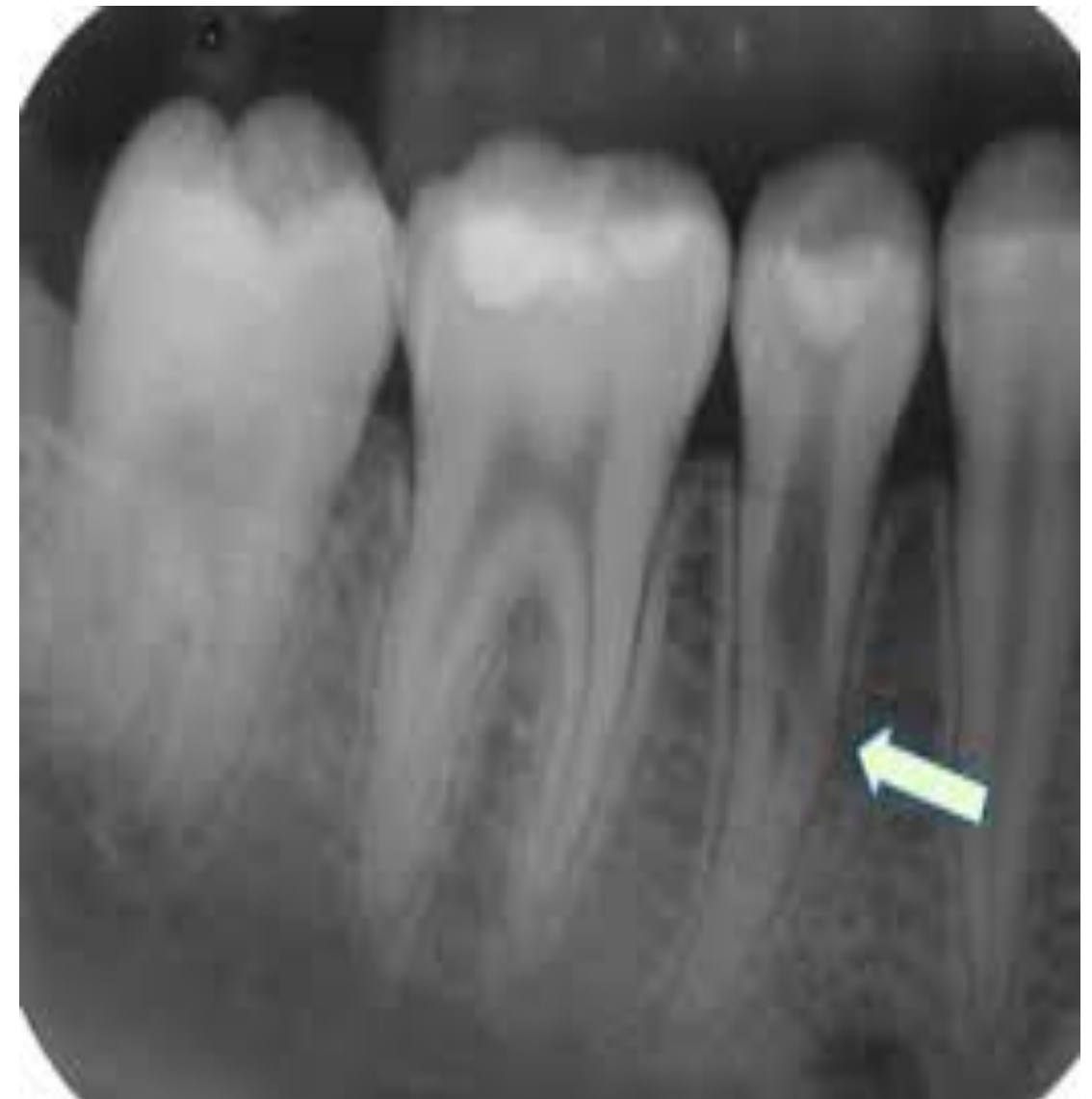
Badly carious tooth, heavily filled crown, non-vital tooth or root canal treated tooth, tooth with curved roots, malposed tooth, roots with external or internal resorption.

➤ Factors related to the investing bone: the surrounding bone might be excessively dense due to localized or systemic causes.

External root resorption



Internal root resorption



➤ Factors related to the operators, include:

1- improper application of the beaks of the dental forceps or elevator on the tooth to be extracted: like the placement of the beaks of the **dental forceps on the crown** instead of the root or below the cemento-enamel junction, also the beaks are **not parallel** to the long axis of the tooth, or the use of wrong type of forceps.

2- **Incorrect application of force during extraction by wrong direction** in addition to that the use of twisting or rotational movement when not indicated.

B- Alveolar bone fracture:

Fracture of alveolar bone frequently occurs when extraction is difficult.

The fractured bone may be removed with tooth to which it is firmly attached or it may be remain attached to the periosteum or it may be completely detached in the socket or wound.

It is a common complication that especially occurs on labial(buccal) area during extraction of upper canine and upper and lower molar teeth.



This complication might be due to:

1. The alveolar bone is very thin
2. Accidental inclusion of the alveolar bone within forceps blades
3. Configuration of the roots
4. The shape of the alveolus.
- 5 Physiological changes in the bone itself like ankylosis.

Tooth ankylosis



Fig. 18.6. Ankylosis; second primary molar.



C. Maxillary tuberosity fracture:

Sometimes the tuberosity is completely fractured when we try to remove maxillary 3 or 2 molar.

Fracture of maxillary tuberosity may lead to a **irregular tearing in the covering soft tissue with bleeding and post-operatively may lead to difficulties in the retention of upper denture.**

This complication might occur if the molar tooth to be extracted is isolated and subjected to full force of **bite leading to sclerosis of the surrounding bone.**



- ***D-Fracture of the adjacent and opposing tooth.***
- Adjacent teeth occasionally may be damaged during extraction procedures, this may include loosening or dislocation or fracture of the teeth.
- This occur mostly due to **careless use of the dental forceps or elevator by wrongfully using the adjacent tooth as a fulcrum.**
- The opposing teeth may be chipped or fractured if the tooth being extracted yield suddenly to **uncontrolled movement of the forceps that strike the opposing tooth leads to this complication.**

E-Mandible fracture:

This is a rare complication, but it might occur almost exclusively with the **surgical removal of impacted lower third molar** tooth.

A mandibular fracture is usually the result of the application of a **force exceeding that needed to remove a tooth** and often occurs during the use of dental elevators(**winters elevator**), but sometimes **pathological or physiological** changes may lead to weakened mandible like:

Senile atrophy, osteoporosis, osteoradionecrosis, osteomyelitis, cystic lesion, impacted teeth and tumors (benign or malignant).

3- Post -operative pain

When a tooth is extracted , the surrounding tissues (bone, gum, ligaments, and nerves) experience trauma.

The body reacts with inflammation and healing processes, which **naturally produce pain** for a few days.

Most people feel mild to moderate pain for about 24–72 hours, and it gradually improves. However, if pain becomes severe, increases after a few days, or lasts longer, it may indicate:

- Traumatized hard tissue resulted from bruising of bone during instrumentation and using burs for removal of bone.
- Damage and rough handling of soft tissue during extraction is another cause for postoperative pain.
- Dry socket

4- Post- operative swelling:

- After extensive surgical interference and exodontias some time may be associated with post-operative swelling, this swelling may be related to one or more of the following:

4- Post- operative swelling:

- **a. Oedema:** oedema occurs after surgery as a result of tissue injury (it is normal response) when there is great damage to the tissue. Using blunt instrument and rough handling of tissue may Increase the chance of production large oedema. So laceration of tissue during extraction, trauma to the bone or periosteum are some of the most common causes of oedema and in other words post-operative swelling.

- **b. Infection:** Swelling due to infection can be distinguished from postoperative oedema by the increased skin temperature, greater redness of the overlying tissues, the usual presence of fever and sometime fluctuation is present due to presence of pus. The infection should be always considered a serious complication and need urgent management.

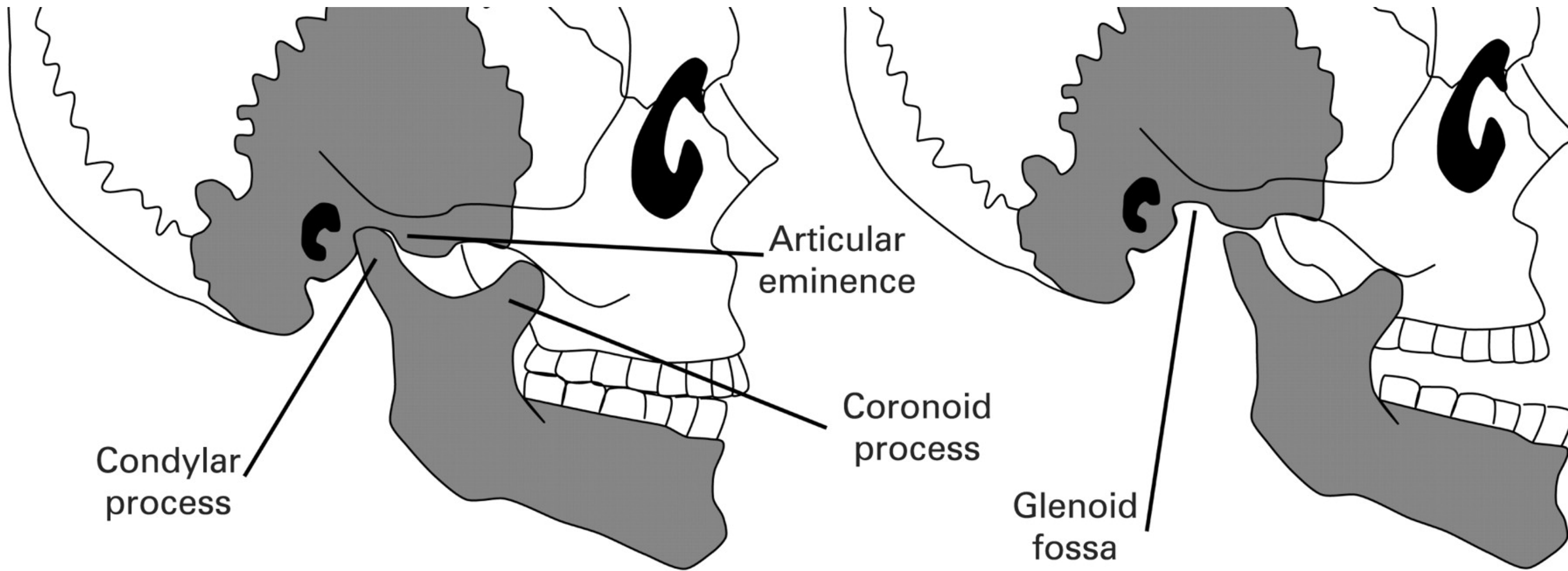
- **c. Hematoma:** A collection of blood in the extra-vascular spaces of the tissues. It is **rare complication** following extraction of the teeth, but sometimes hematoma or ecchymosis (bruising) may develop postoperatively **if haemostasis is not developed and persistent bleeding** from either the socket or adjacent alveolar bone.

5- T.M.J dislocation

- This complication may occur **due to muscle exertion** usually during extraction of mandibular posterior teeth, this may lead to condylar dislocation so the patient became unable to close his mouth.



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Condylar
process

Articular
eminence

Coronoid
process

Glenoid
fossa

Normal

Dislocation

Management

- Reduction is done with the thumb wrapped with gauze to avoid injury by teeth and placed on the occlusal surfaces of mandibular posterior teeth and other fingers under the lower border of the mandible.
- Mandible is then pushed downward backward rotating the chin upwards with this manpower the condyles are moved downwards and backwards over the articular eminences of temporal bone.

- Reduction can be attempted under 5-10mg of IV/IM valium.
- Patient should be warned not to open his mouth too widely and instructed to support the jaw during yawning.
- Extra oral bandage support for the joint is applied and worn until tenderness in the affected joint subsides.



6. Displacement of a root into the soft tissue and tissue spaces and the maxillary antrum:

- A root or piece of root may be dislodged into the soft tissue through a very thin bony plate overlying socket and **disappear buccally or lingually into the soft tissue. E.g**
- **lower molar teeth or roots** may be displaced into the sublingual or submandibular space.
- **upper third molar** may be displaced into the infratemporal space.
- retained root may be displaced into the **antrum** during the extraction of upper molar or sometime premolar teeth especially **palatal root of upper molars.**

- The factors that may assist this complication are :
 1. Use of excessive force without direct vision
 2. Presence of pathological conditions like cysts
 3. The presence of large maxillary antrum.

Displacement of a root into the soft tissue space (submandibular space)



Displacement of a root into the maxillary antrum



Do your best

